

Sysbench benchmark report of MonographDB

Test environment (GCP)

Hardware configuration

This test assesses the scalability of MonographDB using three different cluster configurations:

- 1. T-Shirt S: Single txservice node with low CPU, 16 vCores.
- 2. T-Shirt M: Single txservice node with high CPU, 48 vCores.
- 3. T-Shirt L: Three txservice nodes with high CPU, 48 vCores each.

T-Shirt S to T-Shirt M investigates scale-up capabilities, while T-Shirt M to T-Shirt L evaluates scale-out potential. Traditional databases struggle to increase write IOPS when scaling up due to disk bottlenecks. MonographDB circumvents this issue by enabling separate scaling of the log service, achieving high write IOPS while maintaining lower TCO.

Configuration details are provided below.

T-shirt size: S

Service type	EC2 type	Node count	Disk count
txservice	n2-standard-16	1	1
logservice	n2-standard-8	1	3
cassandra	n2-standard-16	1	1
sysbench	n2-standard-32	1	1

T-shirt size: M

Service type	EC2 type	Node count	Disk count
txservice	n2-standard-48	1	1
logservice	n2-standard-8	1	3
cassandra	n2-standard-16	1	1
sysbench	n2-standard-32	1	1

T-shirt size: L

Service type	EC2 type	Node count	Disk count
txservice	n2-standard-48	3	1
logservice	n2-standard-16	1	6
cassandra	n2-standard-16	1	1

Service type	EC2 type	Node count	Disk count
sysbench	n2-standard-32	1	1

Software version

Service type	Software version
MonographDB	0.3.0
Sysbench	1.0.20_2
OS	Ubuntu20.04

Parameter configuration

MonographDB parameter configuration

T-shirt size: S

```
max_connections=5000
mariadb_thread_pool_size=6
monograph_core_num=6
monograph_node_memory_limit=32000
```

T-shirt size: M

```
max_connections=5000
mariadb_thread_pool_size=16
monograph_core_num=16
monograph_node_memory_limit=96000
```

T-shirt size: L

```
max_connections=5000
mariadb_thread_pool_size=16
monograph_core_num=16
monograph_node_memory_limit=96000
```

Prepare test data

Please install [haproxy](#) on sysbench node.

Run the following command to prepare the test data. Port 3390 is listened by haproxy.

```
sysbench /usr/share/sysbench/oltp_common.lua --
mysql_storage_engine=monograph --tables=10 --table_size=1000000 --mysql-
user=sysbench_test --mysql-password=sysbench123 --mysql-host=127.0.0.1 --
mysql-port=3390 --mysql-db=sbtest --threads=100 --report-interval=10 --
auto_inc=off prepare
```

Run the test

Run the following command to perform the test:

`test_name`'s value includes:
oltp_point_select/oltp_write_only/oltp_update_non_index/oltp_read_only/oltp_read_write `thread_num`'s
value includes: 100/200/500/1000/2000/3000

```
sysbench /usr/share/sysbench/$test_name.lua --
mysql_storage_engine=monograph --tables=10 --table_size=1000000 --mysql-
user=sysbench_test --mysql-password=sysbench123 --mysql-host=127.0.0.1 --
mysql-port=3390 --mysql-db=sbtest --time=60 --threads=$thread_num --
report-interval=10 --mysql-ignore-errors=all --rand-type=special --
auto_inc=off --range_selects=true run
```

MonographDB demonstrates excellent scalability in both read and write scenarios. In read-only workloads, it exhibits linear scalability, allowing for efficient handling of high read volumes. For write-intensive workloads, MonographDB offers impressive single-node write performance and the ability to scale horizontally, ensuring data availability and performance even under heavy write loads.

OLTP_POINT_SELECT Performance

T-shirt size: S

Thread Num	TPS	QPS	95th percentile
100	145709.23	145709.23	0.97
200	147126.59	147126.59	2.11
500	120378.40	120378.40	5.18
1000	113480.04	113480.04	10.27
2000	111825.66	111825.66	19.65
3000	111853.52	111853.52	29.19

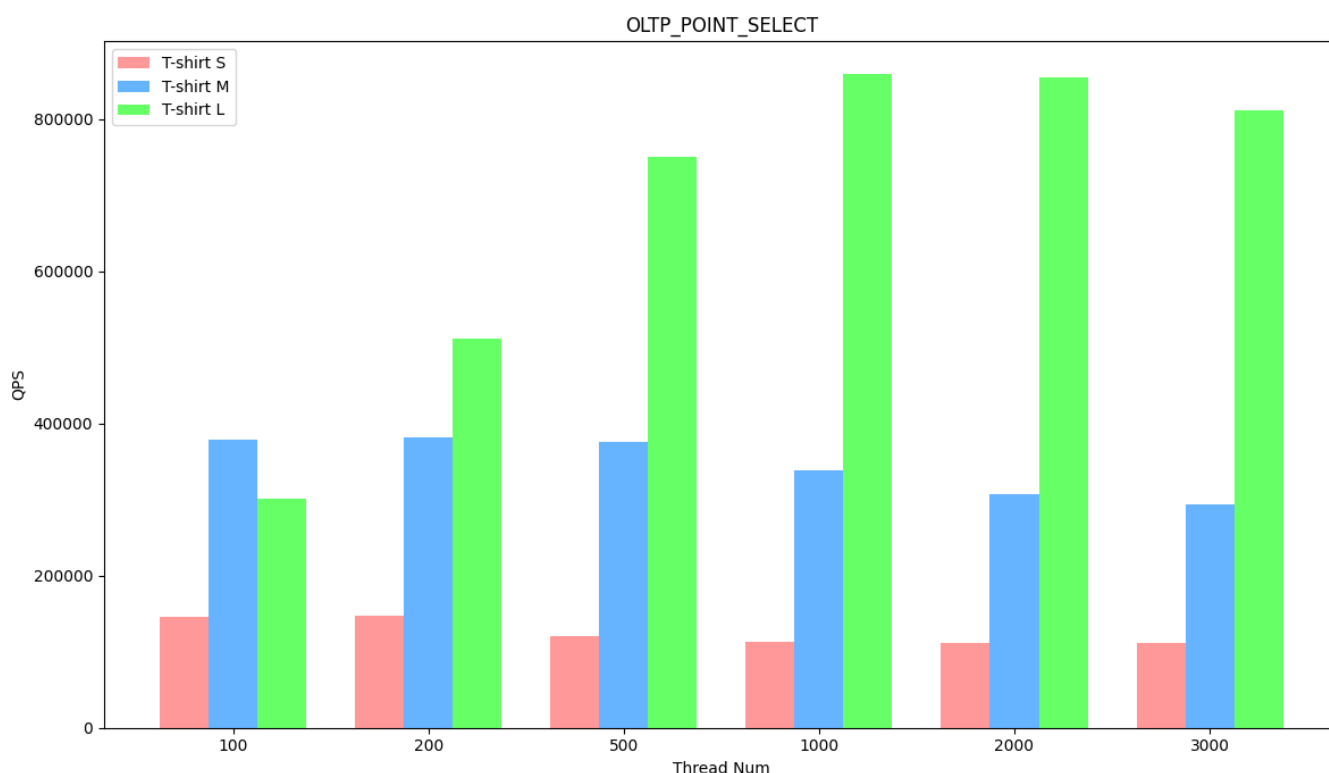
T-shirt size: M

Thread Num	TPS	QPS	95th percentile
100	378984.20	378984.20	0.36

Thread Num	TPS	QPS	95th percentile
200	381346.88	381346.88	0.75
500	376213.56	376213.56	2.11
1000	339065.83	339065.83	3.96
2000	306995.35	306995.35	8.13
3000	293260.17	293260.17	11.87

T-shirt size: L

Thread Num	TPS	QPS	95th percentile
100	300504.05	300504.05	0.50
200	512275.19	512275.19	0.58
500	750484.17	750484.17	1.18
1000	859885.48	859885.48	2.26
2000	854481.67	854481.67	4.65
3000	812597.52	812597.52	6.09



OLTP_WRITE_ONLY Performance

T-shirt size: S

Thread Num	TPS	QPS	95th percentile
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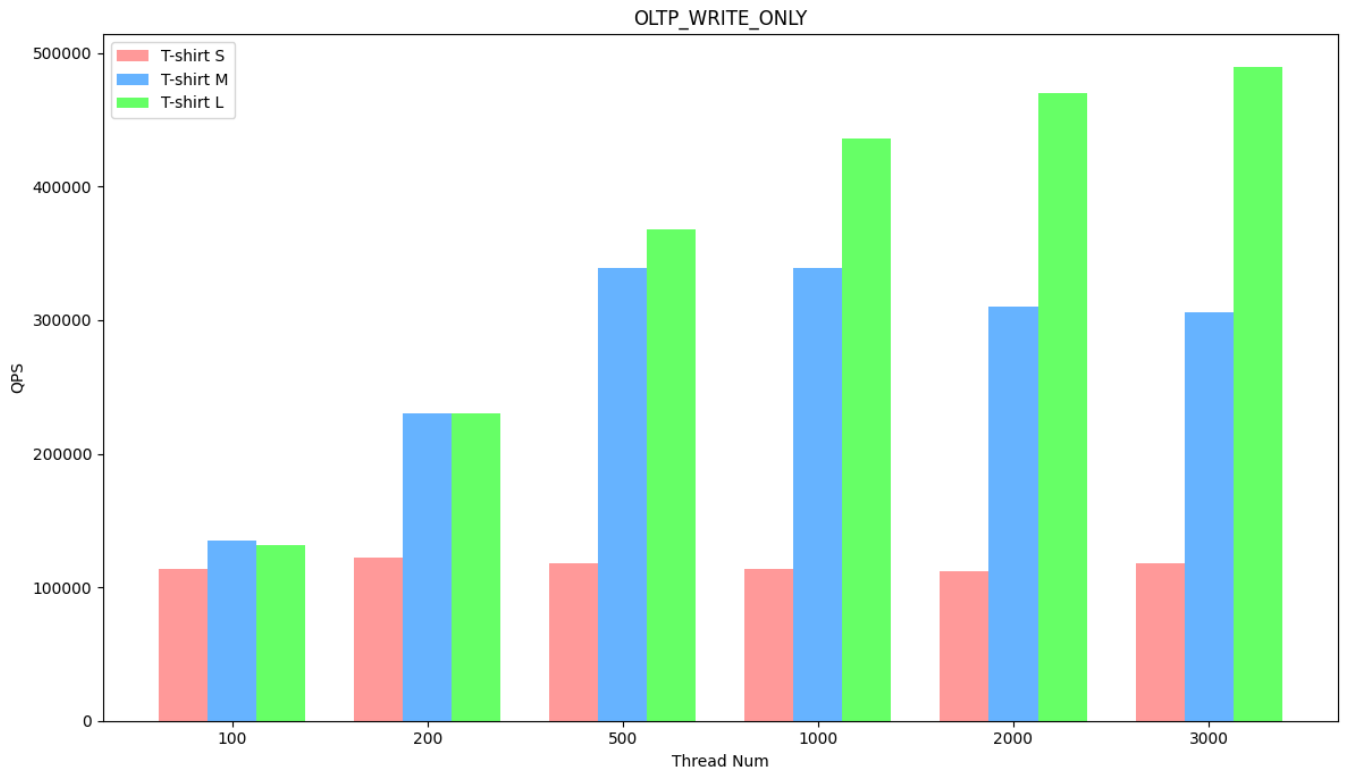
Thread Num	TPS	QPS	95th percentile
100	18969.98	114024.78	6.79
200	20285.62	122059.31	14.21
500	19608.02	118383.20	36.89
1000	18738.20	113850.77	77.19
2000	18228.53	111994.08	158.63
3000	19035.88	118349.57	227.40

T-shirt size: M

Thread Num	TPS	QPS	95th percentile
100	22495.50	135235.11	5.47
200	38246.14	230340.29	6.32
500	55957.52	338667.67	11.65
1000	55699.45	339019.13	23.52
2000	50350.94	309962.83	63.32
3000	49213.04	306280.23	101.13

T-shirt size: L

Thread Num	TPS	QPS	95th percentile
100	21873.79	131499.62	5.47
200	38271.08	230439.25	6.21
500	60839.96	368035.89	11.87
1000	71618.89	436397.47	21.89
2000	76147.14	470420.96	43.39
3000	78195.09	489690.61	62.19



OLTP_READ_WRITE Performance

T-shirt size: S

Thread Num	TPS	QPS	95th percentile
100	3839.92	76893.95	31.94
200	4020.89	80552.95	58.92
500	3999.69	80425.53	150.29
1000	3854.17	77794.61	308.84
2000	3572.60	72579.76	601.29
3000	3393.22	69706.56	926.33

T-shirt size: M

Thread Num	TPS	QPS	95th percentile
100	8199.31	164273.10	17.01
200	9759.38	195694.06	25.74
500	8960.47	179847.43	376.49
1000	9413.23	189506.90	816.63
2000	8619.23	174610.38	419.45
3000	7879.51	159079.63	4280.32

T-shirt size: L

Thread Num	TPS	QPS	95th percentile
100	7410.98	148413.65	16.12
200	11409.70	228829.64	23.52
500	14427.15	289716.38	45.79
1000	16198.52	326014.85	90.78
2000	14696.28	296302.22	240.02
3000	15133.48	305691.68	287.38

